Leibniz's law of continuity, read on records, selects a positive action floor for motions

In the mark model of the Planck paper (§3), the infimum of the error with which a non-adaptive protocol, read by tests invariant under the unknown initial position and velocity, decides Newton's comparison, inertia against a constant force F over a cell of duration τ, is

$$P_*(F) = \Phi\Big(-\tfrac{1}{2}\sqrt{K_\tau/\kappa}\Big), \qquad K_\tau = \frac{F^2\tau^3}{24\,m},$$

for a mark floor $\kappa > 0$, while $P_*(F) = 0$ for every $F \neq 0$ when $\kappa = 0$; at $F = 0$ the cases coincide and $P_* = \frac{1}{2}$. The infimum is approached by dense protocols. **The best verdict over these protocols is continuous in the force at $F = 0$ if and only if $\kappa > 0$**; each single protocol gives a continuous verdict on both branches. The same holds on Zeno's rung, rest against uniform motion, with $mv^2\tau/(8\kappa)$ in place of $K_\tau/(4\kappa)$, attained by two marks. Any finite schedule of marks reaches exactly the cut measure's spent action over κ, so a single cut at fraction s carries $d^2 = 3s(1-s)K_\tau/\kappa$ (cut-measure note, Proposition 7). Static shapes behave differently: when the read coordinate has no dynamics in the window, the best verdict jumps at zero for every κ. The equivalence concerns the observer ignorant of the preparation; a protocol that knows it (Yuen's, in §3 of the paper) decides at every force for every κ by letting its phase-space aperture grow. At a bounded aperture the conclusion returns for that observer too, with $\hbar$ in place of κ (§1, last paragraph).

Leibniz published the law of continuity in July 1687, the year of the *Principia*, as a test of laws of motion: when two cases approach and are lost in one another, their outcomes must do the same. He measures the difference of the cases by discernibility, describing the nearly equal case of a collision as one "dont à peine ce cas peut estre distingué" (§2). Applying the same measure to the outcomes is our step. Read that way, the law holds for Newton's comparison exactly when $\kappa > 0$. Read with outcomes measured by positions and velocities, as Leibniz applied it, it holds on both branches. The fork that every route of this programme reaches (positivity as a premise, the zero branch admissible) is therefore located between a reading Leibniz made and one he did not, and the programme's thesis, that quantization is a consistency condition on laws which Newton's continuum limit missed, has a period form: Leibniz's "preuve ou examen" of laws of motion, applied to their records. His *petites perceptions* (1704) supply the graded, additive discernibility that the records reading needs (§3), and his *Pacidius* (1676) the proportion argument against leaps, which a floor on action meets (§2b). Newton's own Rule III (1713) separates division by reason, certain, from division by the powers of nature, "incertum", to be settled by one experiment (§4b), and his *Opticks* posits the permanent least bodies that the *Pacidius* named, for the sake of lasting natures (§4c). Galileo in 1638 already argues the beginning of fall from the graded vanishing of its record, the records reading's measure of outcomes in practice (§4d). The Newton-age dispute over indivisibles, Galileo, Cavalieri and Guldin between 1621 and 1647, concerned static figures, where the theorem imposes no floor and Newton's limit doctrine is the right answer (§4).

Internal model review (Claude Fable, 2026-09-29). REFINE, applied. The mathematics of Proposition L checked, including the $\kappa = 0$ case and the static case; its equivalence at $F = 0$ needs only Theorem 2's inequality. Fable supplied the exact single-cut share $3s(1-s)K_\tau/\kappa$, now in §1 and §3. Corrections: the non-adaptive, preparation-ignorant hypotheses are stated up front (Yuen's protocol jumps for every κ); the transfer of Leibniz's discernibility from the cases to the outcomes, the reading of his gunpowder exemption (with its counter-reading) and the confinement to motions are marked as ours; "distingué" and "sçauroit" are flagged as reconstructed from the OCR; the Epicurus, al-Nazzam and Galileo wordings of §4 are softened to what the texts say; prior art on statistical distance and on continuity axioms is added. A second pass on §§2b, 4b and 4c (same day): REFINE, applied; page and speaker corrections in the *Pacidius*,

the Kepler scaling corrected to $\lambda r(\lambda^{-3/2}t)$, the "convention" of the floor's value qualified against a fourth constant $(e^2/(\kappa c))$, Corollary 3 read as a confidence bound, the transfer of Rule III from parts of bodies to records marked as ours, and Quaestio 23 of the 1706 *Optice* named.

1. The proposition

Availability premise. For $\kappa > 0$, product-saturating marks at every positive resolution and the unbounded end-mark/dense-protocol limiting family are admitted. At $\kappa = 0$, arbitrarily accurate zero-recoil marks (or an equivalent balanced zero-noise family) are admitted. A lower product bound alone implies none of this availability. Aperture bounds below apply to every hybrid/translated conditional state used by the argument.

Setting. The statistical model of the Planck paper, §3: mark j at time t_j returns the position with Gaussian error of standard deviation δ_j and delivers an independent Gaussian impulse of standard deviation Δ_j; every available mark obeys $\delta_j\Delta_j \geq \kappa$; protocols are non-adaptive and tests are invariant under the unknown y_0, v_0. The optimal invariant test errs with probability $\Phi(-d/2)$, and Theorem 2 of the paper gives $d^2 \leq K_\tau/\kappa$ for every protocol, with a sharp constant.

Proposition L. Let $P_*(F)$ be the infimum of the error over admissible protocols and invariant tests. Then $P_*(F) = \Phi(-\frac{1}{2}\sqrt{K_\tau/\kappa})$ if $\kappa > 0$, and $P_*(F) = 0$ for $F \neq 0$ if $\kappa = 0$. For Zeno's rung (rest against speed v, unknown initial position), $P_*(v) = \Phi(-\sqrt{mv^2\tau/(8\kappa)})$ if $\kappa > 0$ and $P_*(v) = 0$ for $v \neq 0$ if $\kappa = 0$. For a static taper ϑ read at two heights, $P_*(\vartheta) = 0$ for $\vartheta \neq 0$ and every $\kappa \geq 0$, when the read coordinate has no dynamics in the window (a clamped figure, or one of effectively infinite mass). For a single cut, marks at $0, s\tau, \tau$, the best d^2 is exactly $3s(1-s)K_\tau/\kappa$.

Proof. For $\kappa > 0$, $\Phi(-d/2)$ decreases in d, and the supremum of d^2 over protocols is K_τ/κ by the sharpness in Theorem 2; hence the infimum of the error. For $\kappa = 0$, marks with $\Delta_j = 0$ and arbitrary $\delta_j > 0$ are admissible; take three marks at distinct times and weights u with $\sum u_i = \sum u_i t_i = 0$, so that $u^{\mathsf{T}}P = \frac{F}{2m}\sum u_i t_i^2 \neq 0$ while $u^{\mathsf{T}}\Sigma u = \sum \delta_i^2 u_i^2 \to 0$; then $d \to \infty$ (the same holds at fixed δ by repetition, since $\Delta = 0$ turns the motion into a static figure). The equivalence at $F = 0$ needs only Theorem 2's inequality, whose proof is complete: $d^2 \leq K_\tau/\kappa$ gives $P_* \geq \Phi(-\frac{1}{2}\sqrt{K_\tau/\kappa}) \to \frac{1}{2}$; the sharpness fixes only the exact value. Zeno's rung is Theorem 9(i) of the paper, whose bound $d^2 \leq mv^2\tau/(2\kappa)$ two marks attain. The static case is Theorem 9(iii): with N marks of resolution δ at each height, $d^2 = N\vartheta^2\Delta h^2/(2\delta^2)$, unbounded in N whatever κ is. The single cut is the case $N = 2$ of Proposition 7 of the cut-measure note, which treats every finite schedule. □

For $\kappa > 0$ the verdict is Lipschitz at $F = 0$ with constant $\tau^{3/2}/(2\sqrt{2\pi}\sqrt{24m\kappa})$, which diverges as $\kappa \to 0$: the zero branch is the singular limit of a family of continuous verdict functions, the pattern of an order of limits. Here N counts marks on one body inside one window, Newton's single trajectory; repeating the whole experiment on fresh preparations multiplies d^2 on either kind of case. The paper's quantum results (§§4–6) reproduce the split for the observer ignorant of the preparation: a static parameter can be estimated without limit by repeated marks, while the dynamical comparison in a fixed window keeps the N-independent bound. With knowledge of the preparation (Yuen's protocol) a single sharp mark decides at every force, so the proposition is a statement about the preparation-ignorant observer. That observer is Newton's in Book III, who reads forces from the phenomena, motions nobody prepared; §10 of the paper prices the preparation route by the body's spread along the way.

Both observers, at bounded aperture. For the observer who knows the preparation, Theorem 2 of the probabilistic note covers every finite adaptive protocol of instruments, with any Kraus operators and apparatus memory, whose conditional states keep the body within an aperture (L, P): deciding at error ϵ requires $F\tau L/\hbar + F\tau^2 P/(2m\hbar) \geq 1 - 2\epsilon$. Hence, with P_*

now the infimum over protocols of aperture (L, P) and any preparation,

$$P_*(F) \;\geq\; \tfrac{1}{2}\Big(1 - \frac{F\tau L}{\hbar} - \frac{F\tau^2 P}{2m\hbar}\Big) \;\longrightarrow\; \tfrac{1}{2} \qquad (F \to 0),$$

so the best verdict is continuous at $F = 0$ whenever $\hbar > 0$ and the aperture is bounded, while a classical apparatus with sharp preparations decides at every $F \neq 0$. Yuen's protocol escapes only by letting L or P grow as F shrinks, an idealization of the same kind as the perfect instrument. For the preparation-ignorant observer, Theorem R of the recoil note, proved for non-adaptive protocols with any product probe states (constant $1 - 2\epsilon$, sharpened to $\arcsin(1 - 2\epsilon)$ by the path-length note) and extended there to adaptive choices by the hybrid argument with each Δ_j taken as its supremum at step j, gives $s \sum_j \Delta_j \geq 8\hbar \arcsin(1 - 2\epsilon)$, with $s = F\tau^2/(2m)$ the sagitta and Δ_j the impulse spreads; at bounded total recoil that observer's best verdict is again continuous at $F = 0$. The records reading of Leibniz's law therefore holds for both observers exactly on the floor branch, at bounded apertures.

2. Leibniz, 1687: the law of continuity as a test of laws

Leibniz, "Lettre de M. L. sur un principe general utile à l'explication des loix de la nature", *Nouvelles de la République des Lettres*, July 1687, in Gerhardt, *Die philosophischen Schriften* III (1887), 51–55 (companion; passage, OCR normalized by hand; where the OCR misreads accents, the reconstructed letters are flagged).

- **The text.** "Lorsque la difference de deux cas peut estre diminuée au dessous de toute grandeur donnée *in datis* ou dans ce qui est posé, il faut qu'elle se puisse trouver aussi diminuée au dessous de toute grandeur donnée *in quaesitis* ou dans ce qui en resulte ... *Datis ordinatis etiam quaesita sunt ordinata*" (p. 52). "Le repos peut estre consideré comme une vistesse infinement petite, ou comme une tardité infinie", so that the rule of rest must be a particular case of the rule of motion, "autrement ... ce sera une marque asseurée, que les regles sont mal concertées" (pp. 52–53). Against Descartes's second rule of collision, after "une augmentation aussi petite que l'on voudra du corps B auparavant égal à C", B should reflect a little less and C a little more "qu'au cas de l'égalité dont à peine ce cas peut estre distingué" (p. 53; the OCR reads "distinguo", and the final letter awaits the page image). And the exemption: "dans les choses composées quelques fois un petit changement peut faire un grand effect, comme par exemple une estincelle tombant dans une grande masse de la poudre à canon", while "à l'égard des principes ou choses simples, rien de semblable ne sçauroit arriver" (p. 54; OCR "scauroit").
- **What it commits him to.** A law of motion is tested from outside, before any inner discussion ("preuve ou examen ... avant même que de venir à une discussion interieure"), by whether its outcomes converge when its cases do. Rest is the limit of slow motion and equality the limit of inequality, and a rule that jumps at either is ill made. Composite amplifiers may magnify small differences into large effects; simple principles may not jump.
- **What the theorem does with it.** Leibniz applies the test to collision velocities, the geometric reading, and Newton's laws pass it on both branches. His phrase measures the difference *in datis* by discernibility: the nearly equal collision can scarcely be told from the equal one. The records reading applies the same measure *in quaesitis*, and that transfer is ours. It has one textual support: Leibniz individuates things by discernible difference, "deux choses individuelles ne sauroient estre parfaitement semblables" (1704, §3), so results that differ are, for him, results that can in principle be told apart. A second support is an argument of ours from the law's general form, "Datis ordinatis etiam quaesita sunt ordinata": data and results are ordered alike, and measuring the data by discernibility while measuring the results by position mixes two orders. Proposition L shows what the test then demands: for Newton's comparison and for Zeno's rung, continuity of the best

verdict holds exactly when $\kappa > 0$. The exemption marks the boundary of the argument. An instrument is a composite amplifier, and finite amplification is allowed; the zero branch needs the supremum over all instruments the laws permit to jump, a jump in what the laws allow to be recorded, which is the level of principle if the mark trade-off is a law. That mapping of Leibniz's line between composites and simple principles onto the line between one instrument and the laws' supremum is ours, and it has a counter-reading: Leibniz exempts effects for which "on en peut rendre raison par les principes generaux mêmes" (p. 54), and a Leibnizian can explain the $\kappa = 0$ jump from $\kappa = 0$ mechanics by ever finer instruments. Whether the verdict of the best possible record counts among "ce qui en resulte" is the premise, now stated in Leibniz's terms. The theorem also confines the reading: for static shapes the records reading fails on both branches. Leibniz applies the law to geometry and physics alike ("absolument necessaire dans la Geometrie, mais il reussit encor dans la physique", p. 52), so the confinement to motions is a physical choice of ours, resting on back-action; its textual defence is that Leibniz uses the law as a test of laws of nature, and a taper is a datum rather than a law.

Prior art, at metadata level (Crossref verified 2026-09-29 where a DOI is given): Mortensen, "The Leibniz Continuity Condition, Inconsistency and Quantum Dynamics", *J. Philos. Logic* **26** (1997), 377–389, treats the condition against quantum jumps at the instant of change; the pairing of continuity with the identity of indiscernibles is Chapter V of Russell, *A Critical Exposition of the Philosophy of Leibniz* (1900). Leibniz repeats the test against Descartes's rules of collision in the *Animadversiones in partem generalem Principiorum Cartesianorum* (1692) and the *Specimen dynamicum* (1695), so the 1687 "preuve ou examen" is a settled method (recalled, not read here). The condition used here is continuity of the parametrized family of recorded states in statistical distance (Wootters 1981; Braunstein and Caves 1994), in which classical pure states are pairwise perfectly distinguishable. It is weaker than Hardy's continuity axiom, which asks for a continuous reversible transformation between pure states (see the Hardy audit); continuity axioms separate classical from quantum theory also in Masanes and Müller 2011 and Chiribella, D'Ariano and Perinotti 2011.

2b. Leibniz, 1676: little rests, leaps and the animalcula

Leibniz, *Pacidius Philalethi. Prima de motu philosophia*, headed "Scripta in navi qua ex Anglia in Hollandiam trajeci. 1676 Octob.", in Couturat, *Opuscules et fragments inédits* (1903), 594–627 (companion; passage, OCR normalized by hand).

- **The text.** "Ex veteribus Empedocles et ex recentioribus docti quidam Viri quietulas quasdam interspersas asseruêre" (p. 605), because they "capere non potuerunt quomodo motus unus alio celerior esse possit, sine quiete interspersa" (p. 606). Charinus concedes the regress: "tametsi indefinite progrederer subdividendo ac quietulas indefinite exiguas atque indesignabiles, motulis ejusdem naturae miscerem, opus tamen et tempusculis atque lineolis foret" (p. 606), and a rotating radius gives unequal speeds without any rest. A leap is transcreation: the body is "extingui et annihilari, et in B momento post iterum emergere ac recreari" (p. 617). Against it Charinus: "cùm enim magnitudo aut parvitas nihil ad rem faciat", animalcula as much smaller than us as a head is than the earth would find the same absurdity in their leaps, "omnia proportione sibi respondent" (p. 617); and Pacidius: the leaps are "semper ad minora ac minora propelli et nusquam consistere posse in natura rerum ... nulla autem ratio est, cur huic potius quàm illi corpusculorum gradui saltus illi miraculosi ascribantur, nisi atomos scilicet admittamus" (p. 618). Leibniz denies such atoms by the same argument. (Couturat's angle brackets for Leibniz's additions are dropped; the heading "Scripta in navi", p. 594, lies outside the local excerpt and is taken from the companion's metadata.)

- **What it commits him to.** Speed differences need no rests; rests mixed with motions either concede continuous motion or require leaps; a leap of any size needs a privileged grade of bodies, which sufficient reason refuses; hence no leaps, eleven years before the law of continuity.
- **What the theorem does with it.** The little rests are the velocity-switching processes of the telegraph and checkerboard notes: motion at one speed whose state switches at a rate $\omega = mc^2/K$ (there between the two directions; between motion and rest the structure is the same), with the mean speed set by the mixture; with one rate and symmetric switching the mean speed is fixed, so speed differences need state-dependent weights, the signed amplitudes of the checkerboard, which is where Leibniz's "capere non potuerunt" bites. Charinus's regress, rests made indefinitely small among little motions, is the limit $K \to 0$, which converges to continuous motion at the mean speed; the regress is a limit, and its end is the zero branch. The radius answers the motive for the rests and leaves the rate open. Transcreation below the mesh is the paper's reading of al-Nazzam's leap. The animalcula argument is sound against a leap of fixed size: Newton's mechanics has the two-parameter similarity $x \to \lambda x$, $t \to \mu t$ (the dilations of the fifth-postulate note, the zoom of the tangent-groupoid note), and a length breaks it. A floor on action fixes no length of its own; a length appears only with a force and a mass, as the mesh τ_* and its sagitta do. Under $x \to \lambda x$, $t \to \lambda^2 t$ at fixed mass, $F \to F/\lambda^3$ and $K_\tau = F^2\tau^3/(24m)$ is invariant, as is $mv^2\tau$, so animalcula scaled this way reach the same verdicts in the corresponding cases: "omnia proportione sibi respondent" survives on a one-parameter subgroup, the diffusive scaling of quantum paths. The dilemma between atoms and no leaps omits this option. What sufficient reason then asks for is a reason for an action scale. The composition results give one constant for all bodies (rotation composition; $\kappa = mD$ in the stochastic route), so that scale converts units of mass, length and time, and its value is a convention of units so long as no second dimensionful constant forms a pure number with it: Newton's G and c do not, and the charge e of §4c does, through $e^2/(\kappa c)$. The composition results rest on stated premises (interaction between bodies in the rotation note, the composition premises of the stochastic route), and both admit zero. The question sufficient reason leaves is zero or positive, which §1 settles for motions under the records reading of the 1687 law. Leibniz between 1676 and 1704 thus supplies the objection and the reply: proportion forbids a privileged size, which a floor on action does not need, and continuity of outcomes read on discernibility requires the floor.

3. Leibniz, 1704: small perceptions add

Leibniz, *Nouveaux essais*, Préface, written 1703–1705, in Gerhardt V (1882), 47–49 (companion; passage, OCR normalized by hand).

- **The text.** To hear the roar of the sea "il faut bien qu'on entende les parties qui composent ce tout, c'est à dire les bruits de chaque vague ... Car il faut qu'on en soit affecté un peu par le mouvement de cette vague ... autrement on n'auroit pas celle de cent mille vagues, puisque cent mille riens ne sauroient faire quelque chose" (p. 47). "La nature ne fait jamais des sauts ... jamais un mouvement ne naist immediatement du repos ny s'y reduit que par un mouvement plus petit" (p. 49). "En vertu des variations insensibles, deux choses individuelles ne sauroient estre parfaitement semblables" (p. 49).
- **What it commits him to.** Every effect, however small, registers a little; perception is graded and additive; noticeable perceptions come by degrees from insensible ones; distinct individuals always differ, by differences that can be insensible.
- **What the theorem does with it.** The mark model makes the addition exact. The best discernibility of any finite schedule of marks is the Newton cell action its cuts spend, over κ, and each added mark adds exactly its cut's share $3s(1-s)K_{\tau_i}/\kappa$, at every finite

stage (cut measure, Proposition 7, with Theorems 1–2 there; Proposition 4 gives the same decomposition as Cameron–Martin energy). The doctrine has two halves, and they fall differently. "Cent mille riens ne sauroient faire quelque chose" asks that every share be positive, which holds for every $\kappa \geq 0$. "Insensible alone" asks that a share be unable to decide by itself, and that half selects the floor: with $\kappa > 0$ a cut shorter than the mark mesh cannot exhibit the force alone and contributes in the assembly, while with $\kappa = 0$ a single cut decides and records of motion have no insensible parts. The zero branch can still place insensibility in the perceiver, as Leibniz's confusion of created substances does, graded from one substance to another; the stochastic route shows how composition turns graded constants into one ($\kappa = mD$ for every body), and the step from confused perceivers to one universal floor is recorded here as a reading. The maxim on motion from rest is Galileo's passage through all degrees of slowness, stated at the beginning of motion, the regime of Lemma X ("ipso motus initio"); Zeno's rung of Proposition L is its records version.

4. The Newton-age dispute over indivisibles, 1621–1647

Book I's closing scholium prefers limits to indivisibles, "quoniam durior est indivisibilium Hypothesis", without naming the dispute it settles. The primary texts are held in `docs/classics`.

- **Galileo's bowl and cone** (*Discorsi*, Giornata prima, Favaro VIII, 74–76, 78–80; companion). Every horizontal section of the bowl (cylinder minus hemisphere) equals the section of the inscribed cone, so the surfaces are "sempre eguali, e ... diminuendosi sempre egualmente, vadano a terminare l'una in un sol punto e l'altra nella circonferenza d'un cerchio ... perché in questa consequenza sola versa la nostra maraviglia". His conclusion: "questi attributi di maggioranza, minorità ed egualità non convenghino a gl'infiniti". *Commitment:* comparison fails among infinites and indivisibles. *The theorem:* the bowl is Democritus's third (bowl and cone are each a third of the cylinder), a static figure, so Theorem 9(iii) and Proposition L put no floor on it, and Newton's scholium to Lemma XI declines the question that makes the wonder: ultimate ratios are limits of ratios, never ratios of ultimate quantities, so the ratio of sections is 1 at every cut and point and circle are never compared. Leibniz takes the opposite side from Galileo in 1687, without engaging his paradox: equality is an infinitely small inequality, so the attributes extend to the limiting case by continuity.
- **Cavalieri's reply** (letter 2992 of 1634, Opere XVI 136–138; companion). Equal parts are removed "essendo noi arrivati al nullo piano tanto nel cono quanto nella scodella", and "non mi dichiaro di componere il continuo d'indivisibili, ma solo mostro che i continui hanno la proportione delli aggregati di questi indivisibili". *Commitment:* ratios of aggregates of sections, with no claim of composition. *The theorem:* the cut measure keeps exactly this stance, a statement about aggregates of cuts (shares summing to K_τ for every schedule); it composes no continuum, and the mark mesh bounds exhibition while leaving division free.
- **Galileo's wheel** (Favaro VIII, 69–72, same companion). A smaller concentric wheel covers the larger wheel's line with its own sides "con l'interposizione di cento mila spazii vacui traposti", and for circles "sì come i lati non son quanti, ma bene infiniti, così gl'interposti vacui non son quanti, ma infiniti"; Simplicio hears in it "quei vacui disseminati di certo filosofo antico", and Salviati's retort about the one "il quale negava la Providenza divina" points to Epicurus without naming him. *Commitment:* a continuum composed of infinitely many unquantified indivisibles, partly full and partly void, crossed by leaps across the voids. *The theorem:* in 1638 Galileo publishes as consistent a structure that converges with al-Nazzam's leap; no transmission evidence is known, and the entry records convergence. The paper's reading of the leap applies: below the mesh a record is consistent with a leap,

the interval stays divisible, and the crossing is exhibited in finitely many steps.

- **Guldin** (*Centrobaryca* IV, 1641, preface p. 4; companion): "Galileus profecto in eodem Dialogo de Motu locali, disputans de infinito, de proprietatibus finitorum, quas infinitis applicare minime liceat, contra ipsum concludit." (The companion's transcription is read from the page images by one reader.) *Commitment:* Galileo's own principle refutes Cavalieri. *The theorem:* it sides with Guldin and Newton on composition (no least magnitude follows from κ) and with Cavalieri on ratios of aggregates.

The four entries share one feature that sharpens the paper's §9. The seventeenth-century dispute Newton answered concerned static figures, and there the theorem agrees with his answer and imposes nothing. The floor enters only where a record carries back-action, in the lemmas applied to motion, and the dispute about motion is the one he passed over.

4b. Newton, Rule III (1713; 1726): division by reason and by nature

Newton, *Principia*, third edition (London, 1726), Book III, Regula III, pp. 387–389; the commentary is already in the second edition (1713) (companion; passage, OCR normalized by hand).

- **The text.** "Qualitates corporum quae intendi & remitti nequeunt, quaeque corporibus omnibus competunt in quibus experimenta instituere licet, pro qualitatibus corporum universorum habendae sunt" (p. 387), since "quae minui non possunt, non possunt auferri" and the analogy of nature is "simplex ... & sibi semper consona" (p. 388). Then: "partes indivisas in partes minores ratione distingui posse ex mathematica certum est. Utrum vero partes illae distinctae & nondum divisae per vires naturae dividi & ab invicem separari possint, incertum est. At si vel unico constaret experimento ..." (p. 388).
- **What it commits him to.** Qualities that admit no degrees and are found in every body within reach of experiment belong to all bodies, least parts included, because what cannot be diminished cannot be taken away. Division by reason is certain; division by the powers of nature is an empirical question, and one experiment would settle it.
- **What the theorem does with it.** The second commitment is the theorem's own division of labour. Geometry stays divisible (the paper's Proposition 1; the cut measure's shares for every schedule), and the mark mesh concerns what nature's powers can exhibit, which Newton declares uncertain and empirical. His single experiment has a counterpart here: the exhibition of a force by marks confined to a window shorter than τ_* (the paper's Corollary 3). Corollary 3 is a confidence bound, so one such exhibition at confidence $1-\epsilon$ refutes the floor at that κ and lowers it; the passage to $\kappa = 0$ is the same rule-based step Newton takes to "in infinitum". Rule III speaks of the parts of bodies, and carrying division by nature's powers from parts of bodies to the division of a motion's record is our transfer. The first commitment meets the composition results. One constant for all bodies ($\kappa = mD$ in the stochastic route; rotation composition) makes the floor a quality without degrees in Rule III's sense ("quae intendi & remitti nequeunt"), so Rule III carries it from the bodies within reach of experiment to all bodies, and "quae minui non possunt, non possunt auferri" forbids taking it to zero in the least parts. Newton's rules thus make positivity empirical and universalize it once it is found in every body tested, which composition reduces to one body and its interactions; the settlement itself is experimental. This changes an attribution in the paper: the zero branch is the default reading of Book I's geometry, whose scholium (1687) treats vanishing quantities, "diminuendas sine limite", while Rule III (1713) leaves the division of material parts to experiment; its extension from matter to motion is ours.

Newton on distinguishing motions. The Scholium to the Definitions (1726, pp. 9–11, same companion; the passage stands in the first edition, recalled) sorts true from relative motion by what distinguishes them: "Causae, quibus motus veri & relativi distinguuntur ab

invicem, sunt vires in corpora impressae ad motum generandum. Motus verus nec generatur nec mutatur, nisi per vires in ipsum corpus motum impressas" (p. 9); "Effectus, quibus motus absoluti & relativi distinguuntur ab invicem, sunt vires recedendi ab axe motus circularis ... majores vel minores pro quantitate motus" (p. 10); "Motus quidem veros corporum singulorum cognoscere, & ab apparentibus actu discriminare, difficillimum est ... Causa tamen non est prorsus desperata" (p. 11). Newton compares motions by the effects that distinguish them, graded by the quantity of motion, which is the Galileo comparison's question in his words: whether an impressed force acts is decided by its effects. On the zero branch his "difficillimum" is a practical difficulty; on the floor branch it becomes a bound, the mesh below which no record decides at a given confidence.

4c. Newton, Query 31: permanent least bodies

Newton, *Opticks*, fourth edition (London, 1730), Book III, Query 31, verbatim from the Project Gutenberg transcription (companion; passage). The query goes back to Quaestio 23 of the Latin *Optice* (1706), renumbered 31 in the English edition of 1717 (recalled, not read here).

- **The text.** "God in the Beginning form'd Matter in solid, massy, hard, impenetrable, moveable Particles, of such Sizes and Figures ... even so very hard, as never to wear or break in pieces; no ordinary Power being able to divide what God himself made one in the first Creation. While the Particles continue entire, they may compose Bodies of one and the same Nature and Texture in all Ages ... that Nature may be lasting, the Changes of corporeal Things are to be placed only in the various Separations and new Associations and Motions of these permanent Particles."
- **What it commits him to.** Least bodies of fixed sizes and figures, indivisible by any ordinary power, as the ground of the sameness of natures through time; all change is rearrangement and motion.
- **What the theorem does with it.** These are the atoms that the *Pacidius* names as the only reason to ascribe leaps to one grade of bodies, and Newton holds them, so on Leibniz's own terms Newton could place a floor at the atomic grade. The theorem needs no grade (§2b): a floor on action fixes no length, and Newton's particles supply lengths, one per species, with no universal action. His reason for them is stability, sameness of natures in all ages, and that reason does ask for a scale. In the mechanics of the *Principia* a body bound by an inverse-square force has no preferred size (if $r(t)$ is an orbit, so is $\lambda r(\lambda^{-3/2}t)$, Kepler's third law), so identical sizes for bound systems of one kind need a constant beyond Newton's. He supplies hardness by fiat. The relativistic Kepler note shows that finite propagation speed, which Newton accepts (*Opticks* II.iii Prop. XI), turns singular inverse-square binding into an action threshold $|L| > k/c$; the threshold fixes no size, since circular radii still range over $(0, \infty)$ for $|L| > k/c$, and an action constant of the quantum kind fixes sizes dynamically, Bohr's radius $a_0 = \hbar^2/(m_e e^2)$ in Gaussian units being the standard case. Query 31 thus states, as a premise about matter, the need for a scale that the Galileo comparison states as a premise about records. The two premises are independent: Query 31 posits lengths, the Galileo comparison an action, and each text stops at its own. Its "no ordinary Power" agrees with Rule III's "per vires naturae ... incertum": division by nature is bounded by the powers available, the form the floor takes as a law about what nature can exhibit.

4d. Galileo, 1638: the beginning of fall read from its record

Galileo, *Discorsi*, Giornata terza, Favaro VIII, 198–200 (companion; passage, from the Wikisource transcription).

- **The text.** Sagredo: a body falling from rest passes through every degree, "grado alcuno non sia di velocità così piccolo, o vogliamo dir di tardità così grande, nel quale non si sia trovato costituito l'istesso mobile dopo la partita dall'infinita tardità, cioè dalla quiete", which the imagination resists "mentre che il senso ci mostra, un grave cadente venir subito con gran velocità" (pp. 198–199). Salviati answers with a weight dropped onto yielding ground, which sinks four fingers from four braccia ("lo ficca in terra, v. g., quattro dita") and less from each lower height: from a finger's height, "che farà di più che se, senza percossa, vi fusse posto sopra? certo pochissimo: ed operazione del tutto impercettibile sarebbe, se si elevasse quanto è grosso un foglio. E perché l'effetto della percossa si regola dalla velocità del medesimo percuziente, chi vorrà dubitare che lentissimo sia 'l moto e più che minima la velocità, dove l'operazione sua sia impercettibile?" (p. 200). And from reason: since velocity is "augumentabile e menomabile in infinito", no reason makes the body enter ten degrees rather than four, two, one, a half, a hundredth (p. 200).
- **What it commits him to.** The beginning of fall passes through all degrees of slowness; the effect of a percussion is graded by the velocity and vanishes, to sense, for the smallest drops; from an imperceptible effect one may infer a minimal velocity; and sufficient reason forbids a jump into any finite degree.
- **What the theorem does with it.** Galileo argues the continuity of the motion from the graded vanishing of its record, measuring outcomes by how perceptible they are: the records reading's measure of outcomes, used forty-nine years before Leibniz published the law, in the service of the geometric reading's conclusion about the velocity. The fall from rest is Newton's comparison with $F = mg$ and $\tau = \sqrt{2h/g}$, so $K_\tau = m\, g^{1/2}(2h)^{3/2}/24$, and Theorem 2 of the paper (Corollary 3 with the window equal to the whole fall) gives, on the floor branch, a least height below which no record of the fall itself decides at confidence $1-\epsilon$ that the weight was accelerated during the drop rather than moved freely, whatever its unknown initial velocity:

  $$h_* = \tfrac{1}{2}\big(96\, z_{1-\epsilon}^2\, \kappa/(m\, g^{1/2})\big)^{2/3},$$

  about 1.7×10^{-22} m for a kilogram with $\kappa = \hbar/2$ and $z = 2$ ($\epsilon \approx 2.3\%$). Galileo's sheet of paper is a threshold of sense, some eighteen orders of magnitude above this threshold of law for the preparation-blind monitoring of the same fall, and his inference from imperceptible to minimal holds on both branches. His own record, the sinking of the weight, is one terminal reading against a known release; the invariance of Theorem 2 removes exactly such a reading, so his protocol is the preparation-knowing one of §10 of the paper, whose floor is $\theta^2\hbar^2/(4LP)$ at the balanced aperture. His argument from reason is the one the *Pacidius* repeats (§2b), and it rules out a jump into a finite degree of speed; a floor on records leaves that question open.

5. Consequence for STATE

The premise can be stated as one sentence about topologies. On the floor branch the best statistical distance between the recorded motions of a cell, supremal over the admissible protocols (non-adaptive, invariant tests, the preparation-ignorant observer), $d(F, F')^2 = (F - F')^2\tau^3/(24m\kappa)$ by Theorem 2 applied to the difference of forces, is a continuous function of the forces that vanishes only on the diagonal, so records and geometry induce the same topology on the motions. On the zero branch that distance is infinite between any two distinct forces, and records induce the discrete topology. Positivity of the floor is therefore equivalent, in the mark model and for the constant-force family under these hypotheses, to the requirement that records and geometry order the motions alike, which is Leibniz's "Datis ordinatis etiam quaesita sunt ordinata" read with the records as the *quaesita*.

For the Newton goal: for the observer ignorant of the preparation, positivity of the action floor is equivalent in the mark model to the records reading of Leibniz's 1687 law of continuity

for laws of motion (Proposition L), and at bounded apertures the observer who knows the preparation reaches the same conclusion through the probabilistic note. The zero branch keeps the geometric reading, which Leibniz himself applied, so the fork stands; it now lies between a reading Leibniz made and one he did not, with his measure of the cases, “à peine distingué”, and the “insensible alone” half of his doctrine of small perceptions on the side of the floor, and statics showing why the reading must be confined to motions, by a physical choice of ours. A single cut carries exactly the cut measure’s share of the discernibility, a result of the mark model worth citing on its own. Newton’s Rule III places the physical limit among empirical questions and supplies, with the composition results, the rule that universalizes a floor found in every body tested, so the zero branch is the default reading of Book I’s geometry; Newton’s stated physics in Book III leaves the division of material parts to experiment. For the scholion: three Leibniz entries (1676, 1687, 1704), Newton’s Rule III and Query 31, Galileo’s weight on yielding ground (1638), and the Galileo–Cavalieri–Guldin layer are supplied with their three obligations; the paper’s §9 carries a pointer. The written model checks of these sections were internal. The corrected Galileo reading retains that the weight itself sinks into the ground and the protocol is the paper’s §10.